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CSCI/MATH 485
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Assignment 2

1. Introduction

Feature selection is an important step in predictive modeling, as it improves model interpretability, reduces overfitting, and can enhance generalization performance. In this assignment, Recursive Feature Elimination (RFE) was applied to the Diabetes dataset from sklearn.datasets. The goal was to identify the most predictive subset of features for modeling disease progression one year after baseline measurements.

The Diabetes dataset contains 10 standardized predictor variables, including demographic and biochemical measurements (age, sex, BMI, blood pressure, and six blood serum variables s1 - s6). The target variable represents a quantitative measure of disease progression.

The dataset was split into 80% training data and 20% test data. A baseline linear regression model was first trained using all 10 features. Then, a custom implementation of RFE was applied to recursively eliminate the least important features based on coefficient magnitude, while tracking test set R^2 at each iteration.

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641

	target
0	151.0
1	75.0
2	141.0
3	206.0
4	135.0

dtype: float64

```
Target Summary:
 count    442.000000
mean     152.133484
std       77.093005
min       25.000000
25%       87.000000
50%      140.500000
75%      211.500000
max      346.000000
Name: target, dtype: float64
```

2. Baseline Linear Regression Model

A linear regression model was trained using all 10 features. The test set R^2 for the baseline model was 0.4526.

```
Initial Linear Regression Coefficients:
age: 1.7538
sex: -11.5118
bmi: 25.6071
bp: 16.8289
s1: -44.4489
s2: 24.6410
s3: 7.6770
s4: 13.1388
s5: 35.1612
s6: 2.3514
Initial Linear Regression R^2 Test: 0.4526
```

The largest coefficients in magnitude were associated with s1, s5, and bmi, suggesting these variables may have strong influence on predicted disease progression. However, baseline regression does not indicate whether all features are necessary for optimal performance. Therefore, RFE was applied to systematically evaluate feature importance.

3. Implement Recursive Feature Elimination (RFE)

3.1 Methodology

Recursive Feature Elimination works by:

- Fitting a model to the training data.
- Identifying the feature with the smallest absolute coefficient.
- Removing that feature.
- Refitting the model.
- Repeating until the desired number of features remains.
- At each iteration, the test set R^2 was recorded to evaluate how model performance changed as features were removed.

3.2 RFE Performance Results

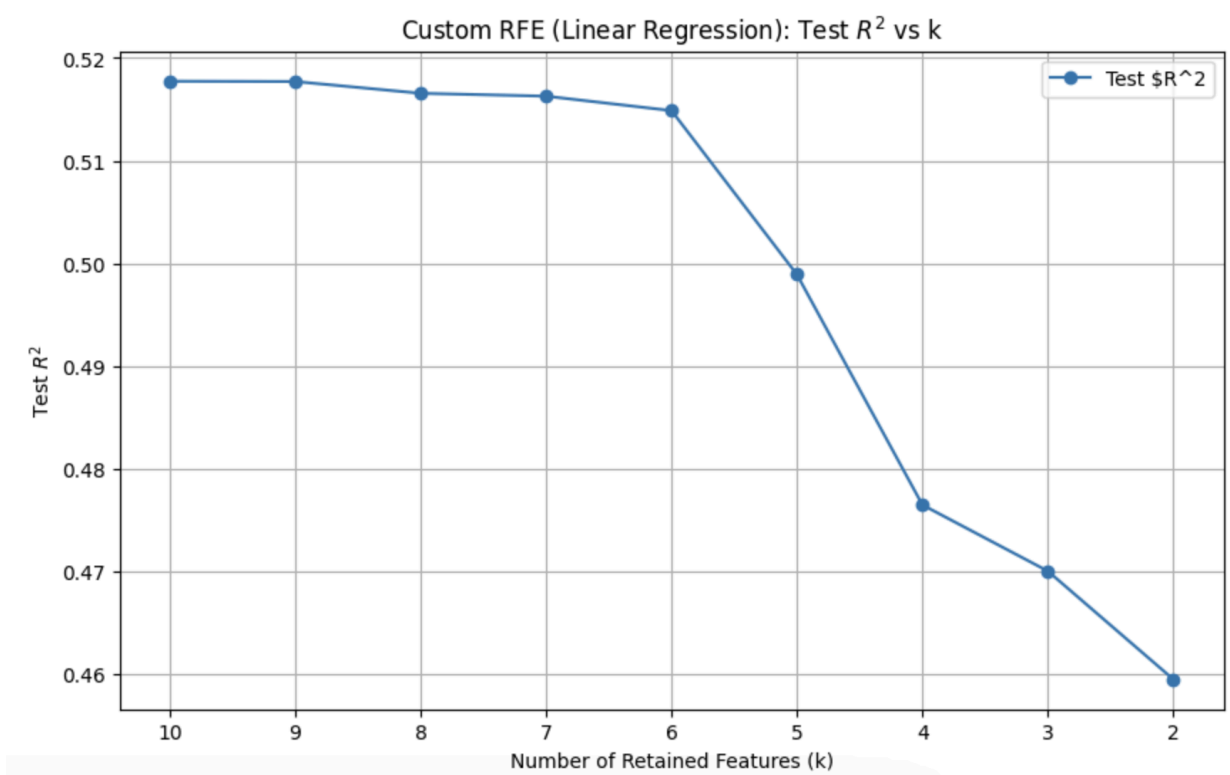
The Test R^2 values across iterations were:

```
Optimal k using threshold =  0.01 => 3
```

	n_features	test_r2	delta_r2
8	2	0.459485	NaN
7	3	0.470025	0.010539
6	4	0.476475	0.006450
5	5	0.498914	0.022439
4	6	0.514884	0.015970
3	7	0.516290	0.001406
2	8	0.516572	0.000282
1	9	0.517717	0.001145
0	10	0.517748	0.000031

The R^2 curve shows that performance remains relatively stable between 10 and 6 features. After fewer than 5 features, performance begins to decline more noticeably.

Using a threshold of 0.01 for performance improvement, the optimal number of features was determined to be k=3. This means that adding additional features beyond three did not improve performance by more than 0.01 in R^2.



4. Optimal Model (k = 3 Features)

At k=3, the selected features and coefficients were:

- s5 = +33.91
- bmi = +32.34
- s1 = −9.23

Interpretation

- **s5 (strong positive coefficient):** This variable has the largest positive influence on predicted disease progression. Higher values of s5 are associated with increased progression.
- **bmi (positive coefficient):** Body Mass Index also has a strong positive relationship with disease progression, which is consistent with medical understanding of diabetes risk factors.
- **s1 (negative coefficient):** This variable has a negative relationship, indicating that higher values are associated with reduced predicted disease progression.

These results suggest that the majority of predictive power in the dataset is concentrated in a small subset of metabolic-related features.

5. Comparison of RFE and LASSO

RFE is a wrapper method, meaning it repeatedly fits a model and removes features based on model-derived importance measures (coefficient magnitude in this case). It explicitly evaluates model performance at each iteration and provides a clear ranking of feature elimination order.

LASSO, in contrast, is an embedded method. It incorporates an L1 regularization penalty directly into the regression optimization process. This penalty shrinks some coefficients exactly to zero, performing feature selection automatically in a single optimization step.

Key differences:

- RFE: Iterative elimination, transparent ranking process, performance tracked at each step.
- LASSO: Single penalized optimization, computationally efficient, automatic shrinkage.

While both methods can produce similar selected features, RFE provides greater interpretability in terms of elimination order and performance behavior as features are removed.

6. Insights About the Dataset

The RFE results indicate:

1. Most predictive signals are concentrated in a small number of features.

2. Several features contribute only marginal improvements to performance.
3. There may be redundancy or correlation among predictors.

The stability of R^2 between 10 and 6 features suggests that multiple variables share overlapping predictive information. However, once the model is reduced to fewer than 4 features, predictive performance declines more noticeably.

This demonstrates that a parsimonious model with three predictors captures most of the meaningful variation in disease progression.

7. Conclusion

Recursive Feature Elimination demonstrated that only three features - s5, bmi, and s1 - are sufficient to retain strong predictive performance in the diabetes dataset. Reducing the model from 10 to 3 predictors maintained competitive generalization performance while improving interpretability and model simplicity.

Overall, RFE effectively identified the most influential predictors and provided insight into how model performance changes as complexity is reduced. This highlights the importance of feature selection in building interpretable and efficient predictive models.

https://github.com/asustayta/MATH-CSCI485_Anna_Sustayta/blob/Assignment2/Assignment2.ipynb

https://colab.research.google.com/drive/1DdzGVeEtmI_FZvpJzlz2ay6TsANyZAMp?usp=sharing